ZIMSEC JUNE 2019 MARKING GUIDE

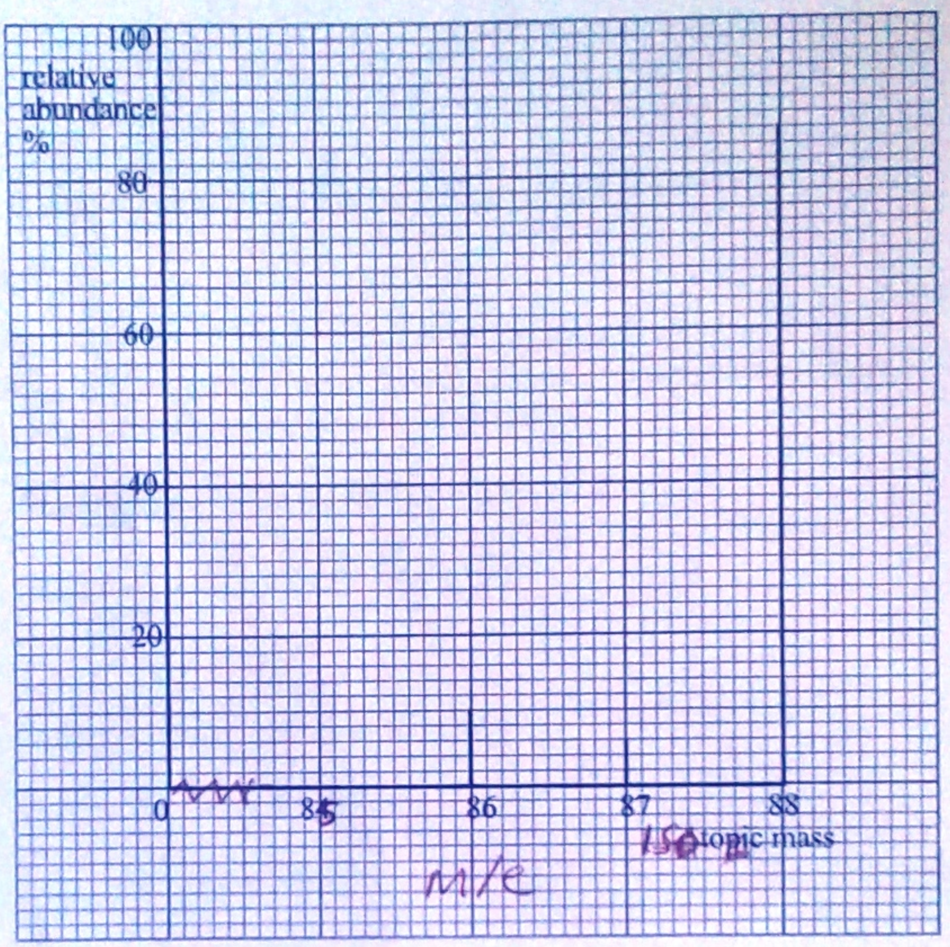
CHEMISTRY PAPER 2

1 (a) (i) – strong electrostatic attraction between a lattice of positive ions and a sea of delocalized electrons/AW [1]

(ii) – number of protons in the nucleus of the atom; [1]

(b) (i) On the grid

Correct abundance peaks [1]
scale and labels [1]



(ii) $= \left(\frac{86\times10}{100}\right) + \left(\frac{87\times7}{100}\right) + \left(\frac{88\times83}{100}\right)$ [1]

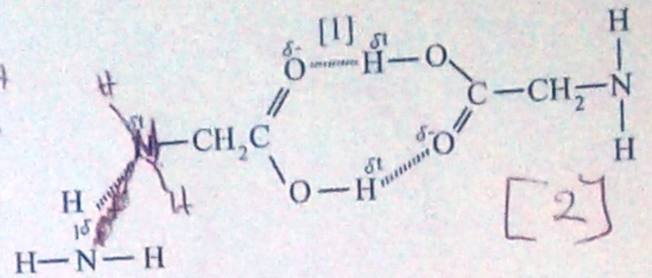
$= 8.6 + 6.09 + 73.04$ [1]

$= 87.73$ R units

(c) - ~~is the attraction between the hydrogen atom of one molecule to the more electronegative atom of another molecule;~~

[1]

- 1 mark for not showing partial charges.



[2]

Accept with water or with NH_3

At least 2 diff.

(d) (i) graphite is slipper due to layers that can slide over each other [1]

weak V.D.W's btwn layers. [1]

[Total: 10]

2 (a) (i) the heat change when one mole of (a substance) an element decomposes completely into gaseous atoms under standard conditions. [1]

(ii) $CH_3CH_2OH_{(g)} \longrightarrow 2\,C_{(g)} + 6\,H_{(g)} + O_{(g)}$ [1]

(iii) $\Delta H\ atm = 5 \times \overset{410}{413} + \overset{350}{347} + \overset{460}{4\,64} + \overset{360}{358} = 3220\ kJmol^{-1}$ [1]

$= \quad +3\,234\ kJmol^{-1}$ [1]

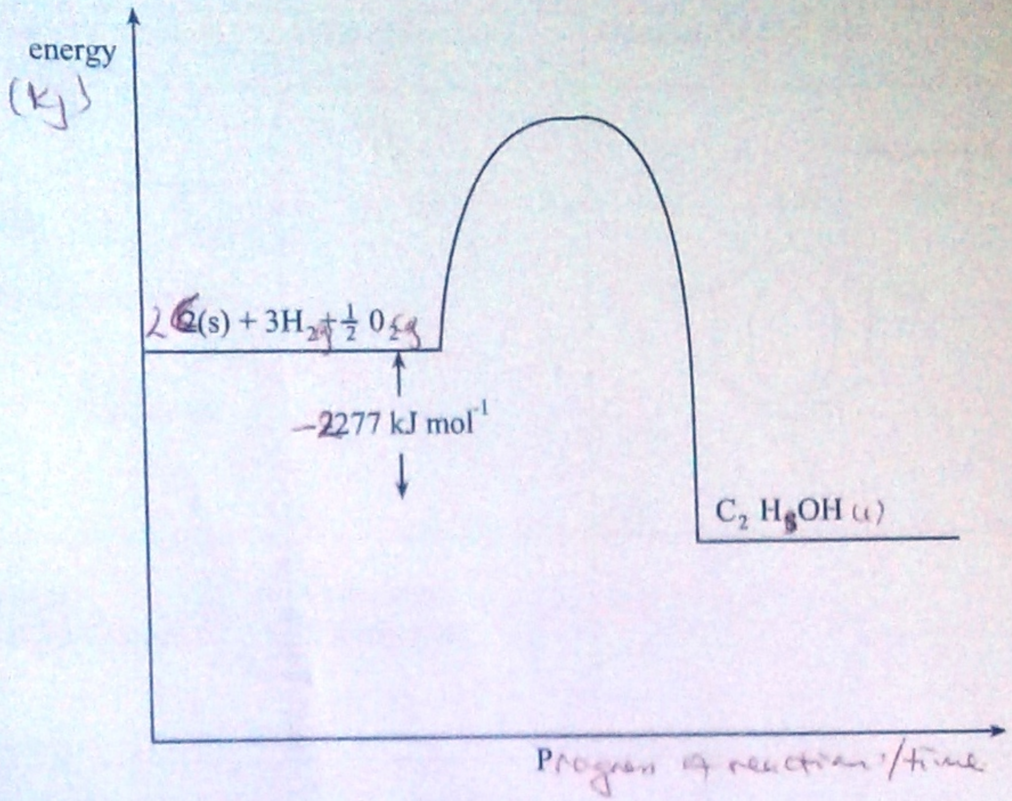
(b) (i) (Standard) enthalpy change of formation [1]

(ii) $\Delta Hr^{\theta} = 2\,(-393) + 3\,(-286) + 1\,367$ [1]

$= -1\,644 + 1\,367$

$= -277\ kJmol^{-1}$ [1]

(iii)



[redacted] labeled axes [1]
Reactants and products labels [1]
ΔH (1) [Total 10]

3 (a) CO_2 - gas [1]

- Simple molecular structure / discrete molecules / weak V.D.Ws btwn particles (intermolecular) [1]

PbO - Solid [1]

- giant ionic structure / strong ionic (electrostatic forces) [1]

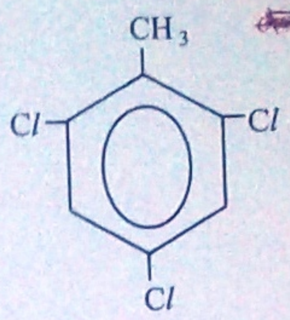
(b) (i) $Ge + 2Cl_2 \longrightarrow GeCl_4$; [1]

(ii) Simple molecular covalent bond; [1]

(iii) low boiling point; [1]
(simple molecule) [redacted]
with weak van der waals forces [1]

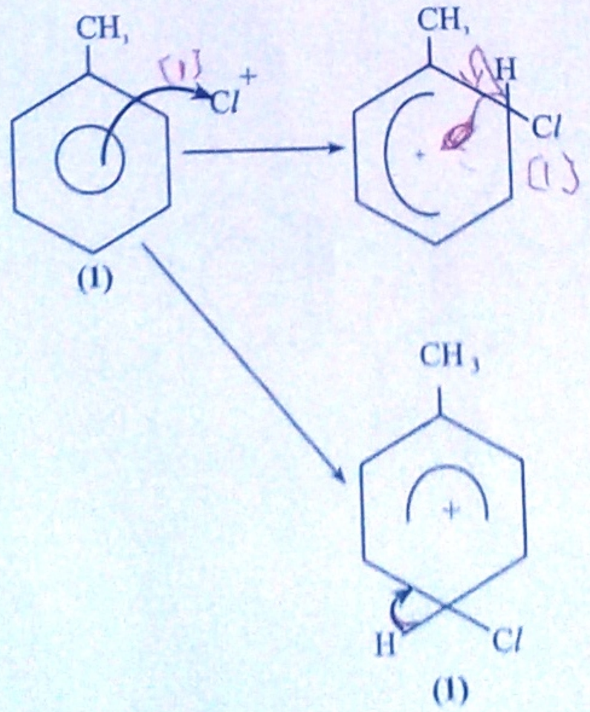
(c) increases down the group; / decreases [1]
~~increase in metallic character;~~ [redacted]
valency electrons more loosely held; / electron (valence) density decreases [1]
[Total 10]

4 (a)

	Conditions required for reaction	Type of reaction	Organic Product (s)
(i)	Fe/ $FeCl_3$ / halogen carrier / $AlCl_3$ / [illegible] Halogen carrier [1]		 [1]
(ii)	~~Boiling methylbenzene;~~ UV light [1]	Free radial; Substitution; [1]	

R-mono
-di

(iii) $Cl - Cl + FeCl_3 \rightleftharpoons Cl^+ + FeCl_4^-$ (1) [1]



~~(b) The electrophile, Cl^+, attacks the benzene ring followed by loss of the H^+ ion.~~

(b) (i) HCl does not dissociate /ionise
to form H^{+}_{aq} ions which make the solution acidic; /A w [1]

They have different intermolecular forces /Aw.

(ii) ~~Too much/insufficient energy required to break H – bonds in water;~~ [1]
~~and the strong forces in methylbenzene~~; [1]

They do not react chemically (1) [Total: 10]

ether /ethoxy.

5 (a) (i) ~~carbonyl~~ /keto; [1]

hydroxyl, [1]

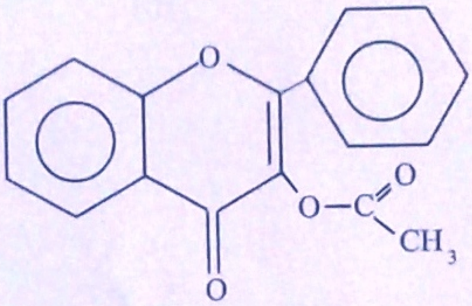
alkene, /Carbon double bond [1]

phenyl aryl /benze any three

(ii) 1. (Bright) orange yellow crystals /ppt [1]

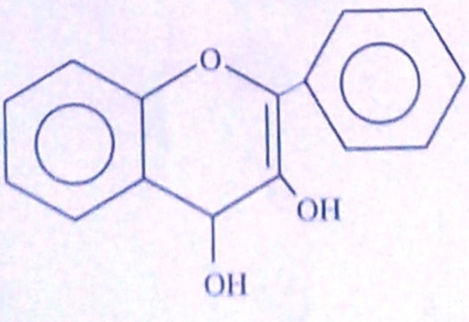
2. aqueous bromine is decolorised /A w [1]

(b) (i)



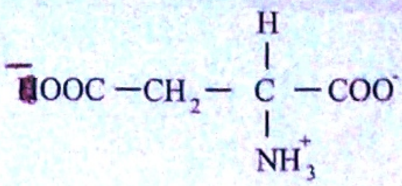
[1]

(ii)



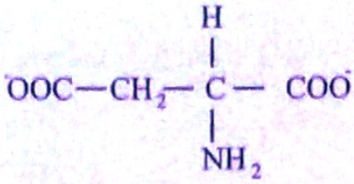
[1]

(b) (i) 1. ~~Zwitterion;~~



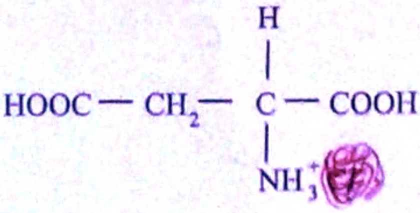
[1]

2. negatively charged ion



[1]

3. +ve ion



R neutral
ionic cmpd.

[1]

[Total: 10]

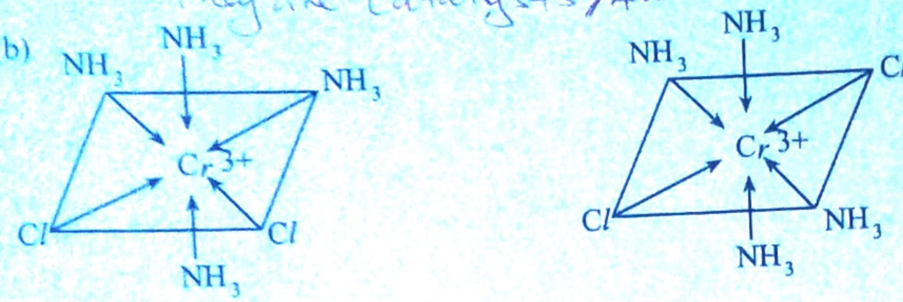
6 (a) (i) Heterogeneous catalysis is a reaction in which the reactants are in different ~~phase with~~ state from that of the catalyst; [1]

(ii) Partially filled ~~vacant~~ 'd'-orbitals; [1]

~~can form variable oxidation states~~; [1]

They are catalysts / AW

(b)



[2]

(c) Study of Science of very small particles; of range 1nm – 100nm; [2]

10^{-9} m – 10^{-7} m

(d) Bioimaging of tissues;
Drug carrier molecules;
Regenerative medicine;
Drug toxicity reduction; [3]

Any 3